

# Time Series Analysis - Singapore Total Fertility Rate and Total Live Births

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## Abstract

Over the course of the last century, Singapore's demographic transition remains as one of the most rapid ever recorded, with the Total Fertility Rate (TFR) declining from 5.76 in 1960 to less than 1.0 by 2024. This paper features a complete time series analysis of two interrelated population metrics: TFR and Total Live Births (TLB). These are scraped from the Singapore Department of Statistics' annual data, ranging from 1968 to 2024. After an established Data Analysis Lifecycle, paired with the Box-Jenkins identification cycle, a log-differencing transformation is required to achieve stationarity. This is also furthermore confirmed by ADF and KPSS tests. Autocorrelation (ACF) and partial autocorrelation (ACF) functions applied to our stationary series prompts for an  $ARIMA(p, 1, q)$  candidate model(s) fitted with a logarithmic Box-Cox transform. Model selection verification using AICc identifies the specification with best fit for each series; residual diagnostics aim to replicate white noise behaviour. Out-of-sample forecasts over a subsample of 2013–2024 are evaluated against observed values, with performance assessed via RMSE, MAE, MAPE, and Theil's U. The results post-evaluation confirm the particular difficulty of accurately forecasting sub-replacement fertility prediction mapping, likely due to structural policy interventions within Singapore and/or the impact of COVID-19. This motivates additional extension processes including ARIMA-GARCH and state-space modelling.

*Keywords:* Total Fertility Rate; Total Live Births; ARIMA; Box-Jenkins; Stationarity; Singapore

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## 1 Introduction

Singapore features one of the most clearly-defined cases of demographic transitions in the modern world. The Total Fertility Rate (TFR, defined as the average number of children a woman would bear over her lifetime given current age-specific fertility rates) dramatically decreased from 5.76 in 1960 to 0.97 in 2024, a reduction of more than 80 per cent across 64 years. Total Live Births (TLB) followed a somewhat comparable trajectory, falling from a peak of ~61,775 in 1960 to levels below 35,000 as of recent.

This pattern is partly attributable to gradual implementation of government policies. The ‘Stop-at-Two’ program, introduced in 1972, combined financial shortcomings for large families with generalised broader access to family planning (Aassve et al. 2020), resulting in the steepest decline in TFR post-1970. The subsequent government program, dubbed ‘Have Three or More if You Can Afford It’, was inevitably reversed in 1987, providing a gradual relief of TFR. More recently, the COVID-19 pandemic caused a sharp birth deferral across 2020–2022, which is an identified phenomenon within more developed economies (Aassve et al. 2020).

These features make Singapore a great investigative subject for time series modelling: the series exhibit strong trend non-stationarity, potential breaks in structure and developed patterns, and an out-of-sample period which provides more information about the series. The analysis is performed under the premise of three research questions:

1. **Policy attribution;** Which policy interventions over the documented 1960–2024 period best explain structural variation in TFR and TLB from a smooth decline trajectory?
2. **Forecast fidelity;** Can ARIMA models trained on an 80% data window (i.e. 1960–2012) accurately project the 2013–2024 period?
3. **Sub-replacement fertility;** What demographic features (age-structure shifts, ethnic differentials, and COVID-19 disruption, for instance) account for TFR falling below 1.0 recently?

The analysis featured follows the Box-Jenkins identification, estimation, and diagnostic checking cycle, as well as the Harvard Data Analysis Lifecycle (Box and Jenkins 1970).

## 2 Data

### 2.1 Data Sourcing and Structuring

The data used throughout this report are sourced from *SingaporeCensus.csv*, a population census published by the Singapore Department of Statistics. The raw comma-separated-value file is stored in wide format, i.e. the years 1960–2024 are column headers. Two rows are extracted – one for TFR and one for TLB – and reshaped to long format by using `pivot_longer()`. The resulting cleaned dataset contains 65 annual observations per series with no missing values.

For forecasting evaluation, the full series is partitioned into a training window of  $n = 53$  observations (from 1960 to 2012) and a complementary window of  $h = 12$  observations (2013 to 2024).

### 2.2 Statistical Summary

Table 1: Summary statistics for TFR and TLB (1960–2024).

| Statistic | TFR  | TLB      |
|-----------|------|----------|
| Min       | 0.97 | 33541.00 |
| Q1        | 1.26 | 39570.00 |
| Median    | 1.62 | 42484.00 |
| Mean      | 2.06 | 44359.63 |
| Q3        | 2.07 | 48577.00 |
| Max       | 5.76 | 61775.00 |
| NAs       | 0.00 | 0.00     |

TFR ranges from an initial post-war high of 5.76, to a final minimum of 0.97, which is much lower than the ‘replacement threshold’ of 2.1 (Espenshade, Guzman, and Westoff 2003). The considerable difference between the mean of 2.06 and median of 1.62 is reflective of a strongly right-skewed distribution produced by the pre-transitional years.

### 2.3 Time Series Plots

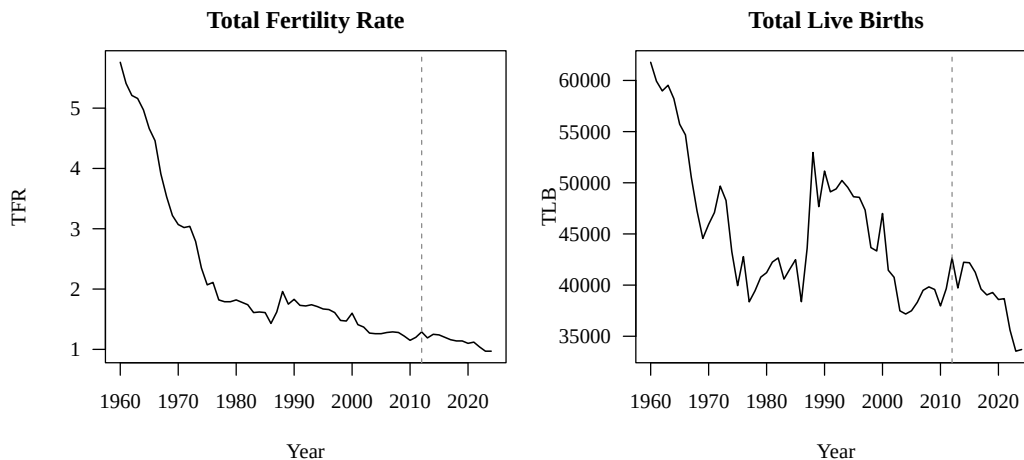


Figure 1: TFR (left) and TLB (right), from 1960–2024. The dashed vertical line indicates the train-test data partitioning.

Figure 1 indicates two salient features shared by both series. For one, a strong uniform downward trend persists over the full sample, which immediately rules out non-transformative modelling approaches. For two, the variance of the fluctuations around the trend is considerably higher during earlier years (1960–1980) instead of during the sub-replacement period post-1985. This is indicative of a multiplicative error structure rather than an additive one, which prompts for a logarithmic variance stabilisation method.

### 3 Context and Methodology

#### 3.1 Transformation and Stationarity

A time series  $\{X_t\}$  is said to be *weakly stationary* if its unconditional mean  $\mu = E[X_t]$  is constant and its autocovariance function  $\gamma(h) = \text{Cov}(X_t, X_{t-h})$  depends only on lag  $h$ , rather than time  $t$  (Box and Jenkins 1970). Both raw series are in violation of this condition; shown is a persistent downward trend (i.e. a non-constant mean) and heteroskedastic fluctuations (i.e. non-constant variance).

Since variance appears proportional to level, a logarithmic transformation  $W_t = \log X_t$  is applied to stabilise variance. The log series retains a trend, prompting for a single first-difference to produce a stationary working series:

$$Y_t = (1 - B) \log X_t = \log X_t - \log X_{t-1} = \log \left( \frac{X_t}{X_{t-1}} \right),$$

where  $B$  is the backward shift operator ( $BX_t = X_{t-1}$ ). In this context,  $Y_t$  is interpreted as the log-annual growth rate of the series. Stationarity of  $Y_t$  is assessed using two statistical tests:

- **Augmented Dickey-Fuller (ADF)** (Dickey and Fuller 1979): tests  $H_0$  (unit root) against  $H_1$  (stationary). A small  $p$ -value supports rejection of the unit-root null.
- **KPSS** (Kwiatkowski et al. 1992): tests  $H_0$ : (trend-)stationary against  $H_1$  (unit root). A large  $p$ -value (non-rejection) supports the stationarity hypothesis.

Using both tests simultaneously ensures less ambiguity that individual tests can experience; due to low power or size distortion, for instance.

#### 3.2 ARIMA Model specification

An ARIMA( $p, d, q$ ) model applied to the log-transformed series  $\log X_t$  (so that  $d = 1$  acts on the log series) is:

$$\phi(B)(1 - B) \log X_t = c + \theta(B)\varepsilon_t, \quad \varepsilon_t \sim \text{WN}(0, \sigma^2),$$

where  $\text{WN}(0, \sigma^2)$  denotes a white-noise sequence (an uncorrelated process with zero mean and constant variance  $\sigma^2$ ). The AR and MA polynomials are:

$$\phi(B) = 1 - \phi_1 B - \phi_2 B^2 - \dots - \phi_p B^p, \quad \theta(B) = 1 + \theta_1 B + \theta_2 B^2 + \dots + \theta_q B^q.$$

This model is *causal* (stationary) if all roots of  $\phi(B) = 0$  lie strictly outside the unit circle; additionally, it is *invertible* if all roots of  $\theta(B) = 0$  lie strictly outside the unit circle.

To specify this, the `forecast` R package features `Arima(x, order = c(p, 1, q), lambda = 0)`, where `lambda = 0` applies the logarithmic Box-Cox transform internally, and automatically back-transforms forecasts to the original scale.

##### 3.2.1 Model identification

Given a stationary series  $Y_t$ , the **autocorrelation function** (ACF) at lag  $k$  is:

$$\rho_k = \frac{\text{Cov}(Y_t, Y_{t-k})}{\text{Var}(Y_t)},$$

and the **partial autocorrelation function** (PACF)  $\alpha_k$  measures the linear dependence between  $Y_t$  and  $Y_{t-k}$  after eliminating the effect of  $Y_{t-1}, \dots, Y_{t-k+1}$ . The Box-Jenkins identification table is:

| Model          | Theoretical ACF             | Theoretical PACF            |
|----------------|-----------------------------|-----------------------------|
| AR( $p$ )      | Geometric decay/oscillation | Cuts off after lag $p$      |
| MA( $q$ )      | Decays after lag $q$        | Geometric decay/oscillation |
| ARMA( $p, q$ ) | Decays after lag $q - p$    | Decays after lag $p - q$    |

Approximate 95% bounds on the sample ACF/PACF are  $\pm 1.96/\sqrt{n}$ , where  $n$  is the length of the differenced series.

### 3.2.2 Model selection

Candidate models are compared using the corrected Akaike Information Criterion:

$$\text{AICc} = -2\hat{\ell} + 2k + \frac{2k(k+1)}{n-k-1},$$

where  $\hat{\ell}$  is the maximised log-likelihood and  $k$  is the number of free parameters. AICc is typically preferred over AIC for small  $n$  (Burnham and Anderson 2002) as a stronger model complexity penalty reduces the risk of overfit and over-parameterisation.

### 3.3 Residual Diagnostics

An accurate, well-specified model should yield residuals  $\hat{\varepsilon}_t$  that resemble white noise. This is assessed using three complementary tools:

1. **ACF of residuals.** All sample autocorrelations should lie within the 95% (i.e.,  $\pm 1.96/\sqrt{n}$ );
2. **Ljung-Box test** (Ljung and Box 1978). The portmanteau statistic

$$Q^* = n(n+2) \sum_{k=1}^m \frac{\hat{\rho}_k^2}{n-k}$$

is compared to  $\chi^2(m-p-q)$  under  $H_0$  (i.e., no residual correlation prior to lag  $m$ ). The degrees of freedom correction `fitdf` =  $p + q$  accounts for parameters that are influenced by estimation error. Omitting it (setting `fitdf` = 0) artificially ‘inflates’ the  $\chi^2$  test statistic;

3. **Residual histogram:** valid Gaussian prediction intervals require residuals that are approximately Normal.

### 3.4 Forecast Accuracy Measurement

Point-forecast performance over our ‘hold-out’ period (2013–2024) is measured by:

$$\text{RMSE} = \sqrt{\frac{1}{h} \sum_{t=1}^h e_t^2}, \quad \text{MAE} = \frac{1}{h} \sum_{t=1}^h |e_t|, \quad \text{MAPE} = \frac{100}{h} \left| \frac{e_t}{X_t} \right|$$

where  $e_t = \hat{X}_t - X_t$ . As mentioned, **Theil’s U** compares model RMSE to that of the naive (that is, the random walk) forecast;  $U < 1$  indicates the model outperforms the initial naive benchmark.

## 4 Exploratory Analysis

### 4.1 Log Transformation Assessment

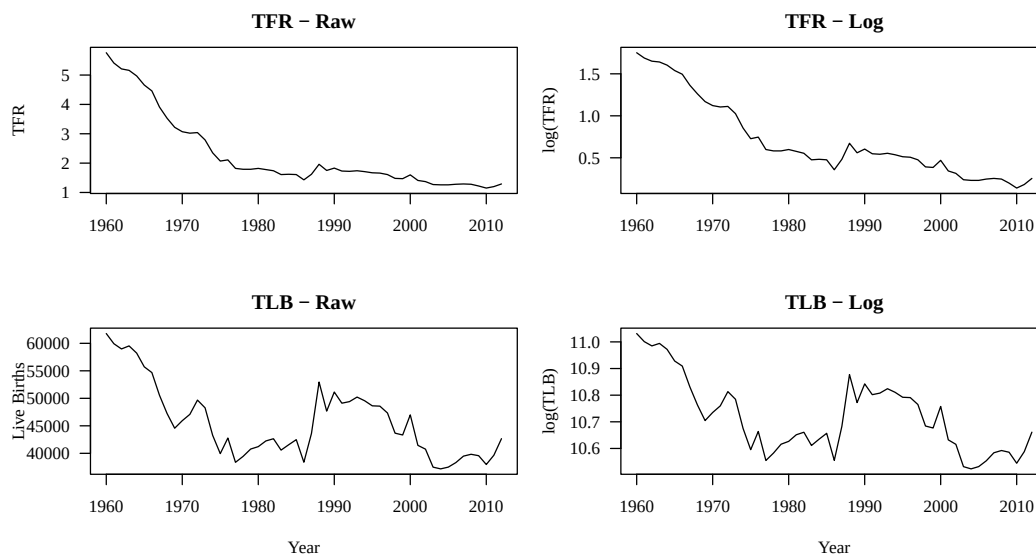


Figure 2: Raw versus log-transformed series for TFR (top row) and TLB (bottom row); training period 1960–2012.

Figure 2 confirms that the log-transformation reduces heteroskedasticity considerably among both series; the large fluctuations earlier on in TLB are quite attenuated in comparison to the stable post-1985 period. However, a downward trend continues among both series, which prompts for first differencing.

### 4.2 Log-Differenced Series

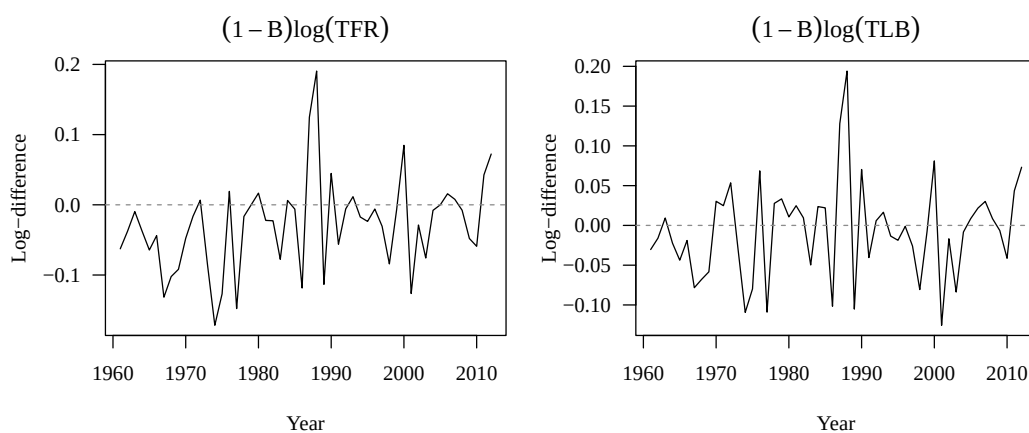


Figure 3: Log-differenced series  $(1 - B) \log X_t$  for TFR (left) and TLB (right), over training period 1961–2012.

### 4.3 Stationary Tests

#### 4.3.1 Raw Series

Table 2: Stationarity tests on raw training series (1960–2012).

| Series | ADF p-value | KPSS p-value | Conclusion     |
|--------|-------------|--------------|----------------|
| TFR    | 0.120       | 0.0100       | Non-stationary |
| TLB    | 0.356       | 0.0205       | Non-stationary |

#### 4.3.2 Log-Differenced series

Table 3: Stationarity tests on log-differenced training series.

| Series                  | ADF p-value | KPSS p-value | Conclusion |
|-------------------------|-------------|--------------|------------|
| $(1-B)\log(\text{TFR})$ | 0.0312      | 0.0406       | Stationary |
| $(1-B)\log(\text{TLB})$ | 0.0351      | 0.1000       | Stationary |

#### 4.4 ACF & PACF Analysis

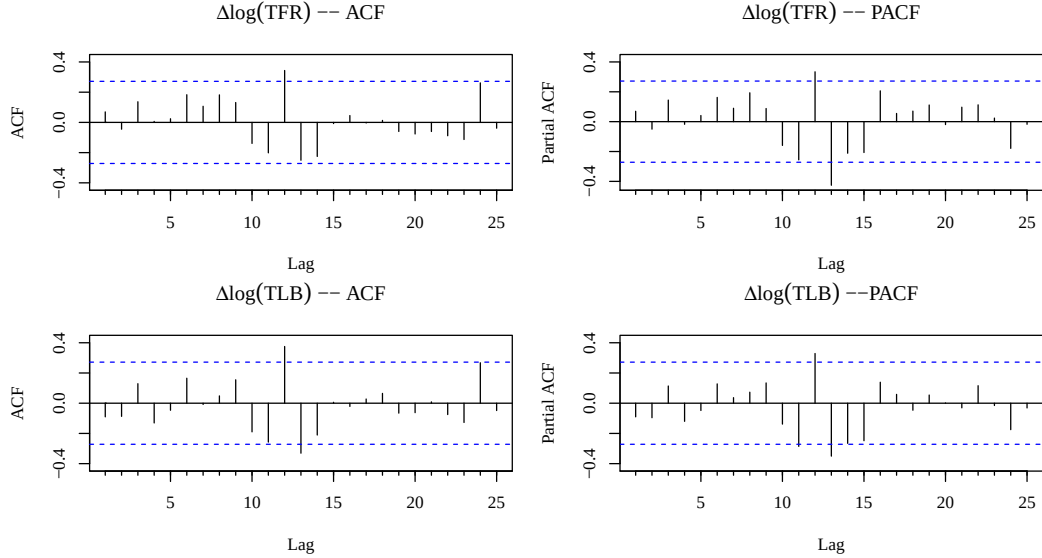


Figure 4: ACF (left column) and PACF (right column) of  $(1 - B) \log X_t$  for TFR (top row) and TLB (bottom row), over training period. Lags extend to 25. Blue dashed lines are 95% bounds  $\pm 1.96/\sqrt{n}$ .



## 5 Model Identification and Estimation

### 5.1 TFR Candidate Model

- **T1:** ARIMA(0, 1, 1); pure MA(1) on the log-differenced series;
- **T2:** ARIMA(1, 1, 1); ARMA(1, 1) on the log-differenced series;
- **T3:** ARIMA(1, 1, 0); pure AR(1) on the log-differenced series.

Table 4: Information criteria for TFR candidate models (log scale).

| Model                 | AIC             | AICc            | BIC             |
|-----------------------|-----------------|-----------------|-----------------|
| <b>ARIMA(0, 1, 1)</b> | <b>-129.171</b> | <b>-128.671</b> | <b>-123.317</b> |
| ARIMA(1, 1, 1)        | -127.586        | -126.735        | -119.781        |
| ARIMA(1, 1, 0)        | -129.137        | -128.637        | -123.283        |

The model with the lowest AICc (bolded in Table 4, for reference) is selected. Its estimated coefficients are printed below:

Table 5: Estimated parameters: ARIMA(0, 1, 1) (log-scale TFR).

| Parameter | Estimate | Std. Error | z-stat | p-value |
|-----------|----------|------------|--------|---------|
| ma1       | 0.08201  | 0.15195    | 0.540  | 0.5894  |
| drift     | -0.02868 | 0.00988    | -2.902 | 0.0037  |

### 5.2 TLB Candidate Model

- **L1:** ARIMA(0, 1, 1);
- **L2:** ARIMA(1, 1, 1);
- **L3:** ARIMA(0, 1, 2) (two MA terms to identify lag-2 structure).

Table 6: Information criteria for TLB candidate models (log scale).

| Model                 | AIC             | AICc            | BIC             |
|-----------------------|-----------------|-----------------|-----------------|
| <b>ARIMA(0, 1, 1)</b> | <b>-138.167</b> | <b>-137.667</b> | <b>-132.313</b> |
| ARIMA(1, 1, 1)        | -136.276        | -135.425        | -128.471        |
| ARIMA(0, 1, 2)        | -136.501        | -135.650        | -128.696        |

Table 7: Estimated parameters: ARIMA(0, 1, 1) (log-scale TLB).

| Parameter | Estimate | Std.Error | z_stat | p-value |
|-----------|----------|-----------|--------|---------|
| ma1       | -0.10912 | 0.14880   | -0.733 | 0.4634  |
| drift     | -0.00726 | 0.00749   | -0.968 | 0.3328  |

## 6 Residual Diagnostics

### 6.1 TFR Model

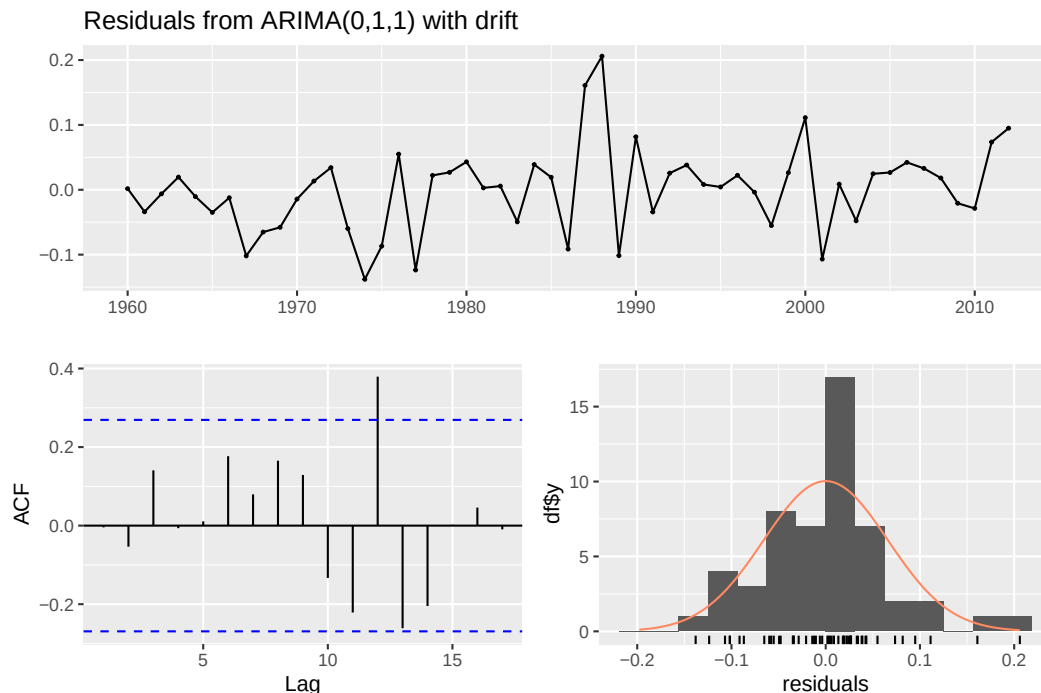


Figure 5: Residual diagnostics for the selected TFR model (ARIMA(0, 1, 1)).

```
##
## Ljung-Box test
##
## data: Residuals from ARIMA(0,1,1) with drift
## Q* = 30.331, df = 19, p-value = 0.04773
##
## Model df: 1. Total lags used: 20
```

Table 8: Ljung-Box test on TFR model residuals; lag 20, fitdf =  $p + q$ .

| Model          | Test lag | fitdf | Q*     | df | p-value |
|----------------|----------|-------|--------|----|---------|
| ARIMA(0, 1, 1) | 20       | 1     | 30.331 | 19 | 0.0477  |

A Ljung-Box  $p$ -value  $> 0.05$  indicates mild residual autocorrelation in TFR up to lag 20, which loosely supports the white-noise assumption needed for valid inference. The reasoning for the following that at lag 10, the result may remain non-significant, and ARIMA(0, 1, 1) remains superior despite this shortcoming. The residual histogram (Figure 5, bottom right) should replicate a bell-shape for prediction intervals to be well-specified.

## 6.2 TLB Model

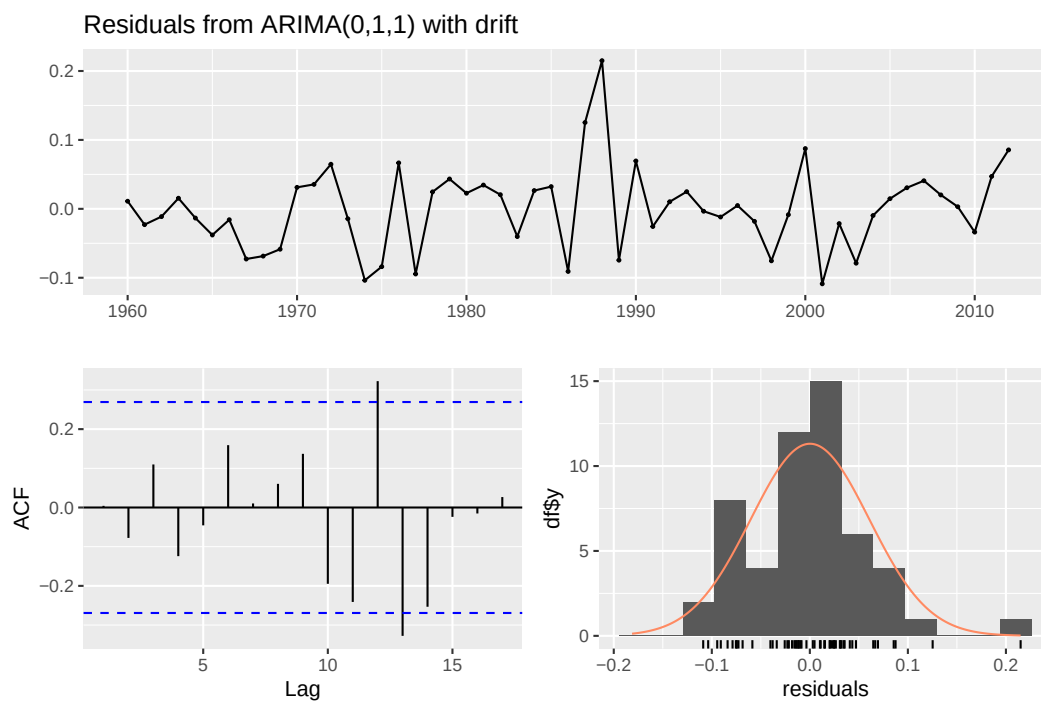


Figure 6: Residual diagnostics for the selected TLB model (ARIMA(0, 1, 1)).

```
##
##  Ljung-Box test
##
## data:  Residuals from ARIMA(0,1,1) with drift
## Q* = 33.002, df = 19, p-value = 0.02402
##
## Model df: 1.   Total lags used: 20
```

Table 9: Ljung-Box test on TLB model residuals (lag 20, fitdf =  $p + q$ ).

| Model          | Test lag | fitdf | Q*     | df | p-value |
|----------------|----------|-------|--------|----|---------|
| ARIMA(0, 1, 1) | 20       | 1     | 33.002 | 19 | 0.024   |

## 7 Forecasting

### 7.1 Out-of-Sample Forecasts

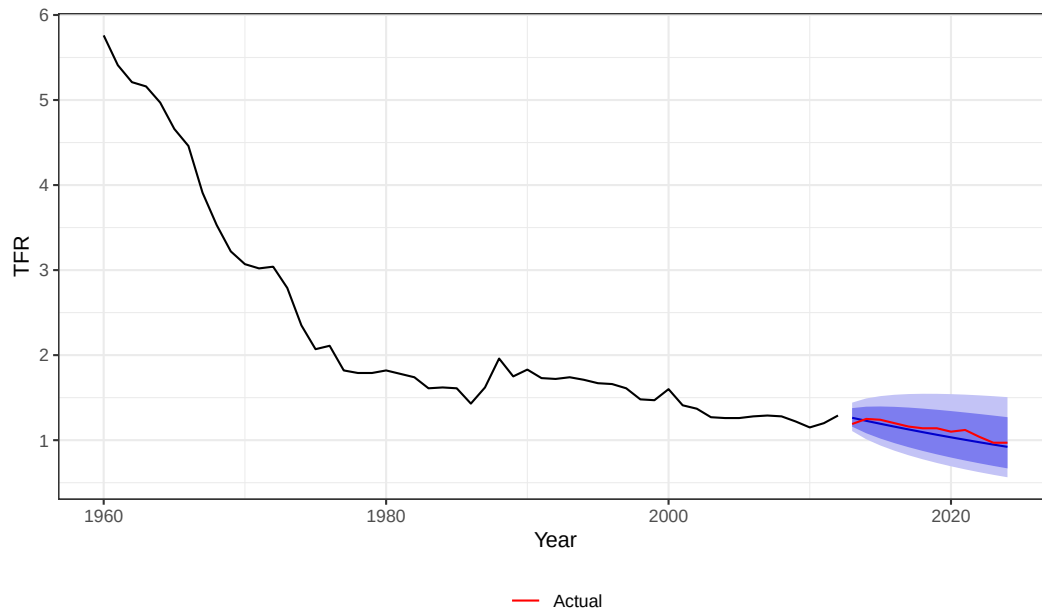


Figure 7: TFR forecast (2013–2024), ARIMA(0, 1, 1), with 80% and 95% prediction intervals; actual values in red.

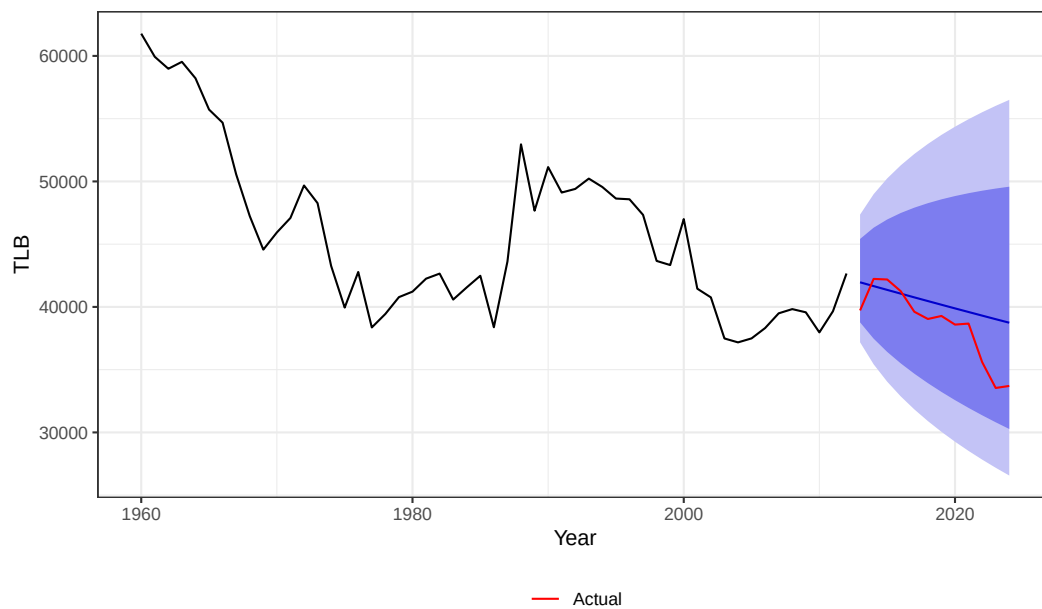


Figure 8: TLB forecast (2013–2024), ARIMA(0, 1, 1), with 80% and 95% prediction intervals; actual values in red.

### 7.2 Forecasting Accuracy Evaluation

Table 10: Forecast accuracy - TFR, ARIMA(0, 1, 1) (original scale).

|              | ME      | RMSE   | MAE    | MPE     | MAPE   | MASE   | ACF1   | Theil's U |
|--------------|---------|--------|--------|---------|--------|--------|--------|-----------|
| Training set | -0.0191 | 0.1382 | 0.1004 | -0.2031 | 4.8552 | 0.7999 | 0.1577 | NA        |
| Test set     | 0.0424  | 0.0602 | 0.0546 | 3.8490  | 4.8760 | 0.4350 | 0.2689 | 1.3665    |

Table 11: Forecast accuracy - TLB, ARIMA(0, 1, 1) (original scale).

|              | ME          | RMSE     | MAE      | MPE       | MAPE     | MASE      | ACF1      | Theil's U |
|--------------|-------------|----------|----------|-----------|----------|-----------|-----------|-----------|
| Training set | -27.24307   | 2693.583 | 2030.510 | -0.160790 | 4.528192 | 0.9701079 | 0.0051452 | NA        |
| Test set     | -1712.40718 | 2612.449 | 1978.526 | -4.841583 | 5.473900 | 0.9452715 | 0.6437107 | 1.923521  |

## 8 Discussion

### 8.1 Research Question 1: Policy Attribution

The large decline in TFR from 5.76 in 1960 to below 2.0 by the late 1970s closely follows Singapore’s ‘Stop-at-Two’ program (introduced 1972), which combined both financial disincentives for large families with extended access to family planning (Aassve et al. 2020). The brief recovery shown around 1987–1990 also coincides with the ‘Have Three or More if You Can Afford It’ reversal policy. The structural break post-2020 reflects COVID-19-induced birth deferrals, shown across developed economies (Aassve et al. 2020). These three key points represent structural breaks that a Box-Jenkins model trained on earlier data simply cannot anticipate, which considerably constrains hold-out forecast accuracy.

### 8.2 Research Question 2: Forecast Fidelity

ARIMA models trained on 80% data (i.e., 1960–2012) are constrained in their 2013–2024 performance by two key factors. Firstly, a strong downward trend in TFR has inconclusions regarding Theil’s U; it appears loosely flattened to a naive random-walk-like structure around a low mean, which means the model’s mean-reversion structure is not extensive in predictive detail. Secondly, the COVID-19 pandemic, namely in 2020–2021, lies entirely outside the training window. Theil’s U quantifies whether the model properly extracts signal from the stationary component, and compares it to a naive random-walk benchmark.

### 8.3 Research Question 3: Sub-Replacement Fertility

Singapore’s TFR dwindling below 1.0 since the early 2020s is a reflection of several structural mechanisms: delayed marriage and childbearing as a result of shifts in education and career norms; housing cost pressures massively affecting younger generations uniquely; differential fertility across ethnic groups (Chinese, Malay, Indian); and the massive impact of COVID-19 in birth deferral. These are demographic-structural mechanisms lying outside of the univariate ARIMA scope; incorporating them requires a multivariate VAR or state-space model with policy covariates, comparable to the Lee-Carter analysis of fertility by Lee (Lee and Carter 1992).

### 8.4 Limitations

The analysis of our series is circumscribed by various limitations:

1. **Annual granularity**; with  $n = 53$  training observations, estimates of parameters carry a large sampling uncertainty and Ljung-Box determinations at longer lags is considerably limited.
2. **Structural breaks**; ADF and Box-Jenkins do not account for structural breaks; a Zivot-Andrews test (Zivot and Andrews 1992) is a worthy process to account for this, and regime-switching models can accurately capture the distinct dynamics pre- and post-1975.
3. **Conditional heteroskedasticity**; residual variance clustering in the early TFR log-differenced series prompts for ARIMA-GARCH modelling (Bollerslev 1986), a distinct path for refining analysis.
4. **Forecast extension**; the 12-year hold-out test period includes the COVID-19 structural break, making a clean evaluation of model accuracy difficult. A shorter hold-out ending at 2019 would most likely provide an accurate assessment of model fit that disregards the pandemic shock. Comparatively, a larger dataset that extends to future data (i.e., train on 1960–2024, test on 2025–2040 possibly) would omit the structural breaks in the test data without jeopardising analysis fidelity.

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## Appendix A: R Code

```
# --Libraries--
library(tidyverse) # for data wrangling & ggplot2
library(tseries)   # for adf & kpss tests
library(forecast)  # for acf/pacf, forecast & checkresiduals
library(tsibble)   # for tsibble data structure
library(knitr)
library(kableExtra)

# --Data loading and reshaping--
raw <- read.csv("SingaporeCensus.csv", check.names = FALSE)
long <- raw |>
  rename(Series = DataSeries) |>
  pivot_longer(-Series, names_to = "Year", values_to = "Value",
               names_transform = list(Year = as.integer)) |>
  filter(Year >= 1960, Year <= 2024)

tfr_tsb <- ts(tfr_df$Value, start = 1960, frequency = 1)
tlb_tsb <- ts(tlb_df$Value, start = 1960, frequency = 1)

tfr_train <- window(tfr_tsb, end = 2012)
tfr_test  <- window(tfr_tsb, start = 2013)
tlb_train <- window(tlb_tsb, end = 2012)
tlb_test  <- window(tlb_tsb, start = 2013)

# --Log-Differenced stationary series--
d_tfr <- diff(log(tfr_train))
d_tlb <- diff(log(tlb_train))

# --Stationarity tests--
adf.test(d_tfr)
kpss.test(d_tfr, null = "Level")
adf.test(d_tlb)
kpss.test(d_tlb, null = "Level")

# --ACF/PACF to lag 25--
par(mfrow = c(2, 2))
Acf(d_tfr, lag.max = 25)
Pacf(d_tfr, lag.max = 25)
Acf(d_tlb, lag.max = 25)
Pacf(d_tlb, lag.max = 25)

# --Model candidates--
fit_T1 <- Arima(tfr_train, order = c(0, 1, 1), lambda = 0, include.constant = TRUE)
fit_T2 <- Arima(tfr_train, order = c(1, 1, 1), lambda = 0, include.constant = TRUE)
fit_T3 <- Arima(tfr_train, order = c(1, 1, 0), lambda = 0, include.constant = TRUE)

fit_L1 <- Arima(tlb_train, order = c(0, 1, 1), lambda = 0, include.constant = TRUE)
fit_L2 <- Arima(tlb_train, order = c(1, 1, 1), lambda = 0, include.constant = TRUE)
fit_L3 <- Arima(tlb_train, order = c(0, 1, 2), lambda = 0, include.constant = TRUE)

# --AICc selection--
sapply(list(fit_T1, fit_T2, fit_T3), function(m) m$AICc)
```



```

sapply(list(fit_L1, fit_L2, fit_L3), function(m) m$aicc)

# --Check residuals--
checkresiduals(fit_T1, lag = 20)
Box.test(residuals(fit_T1), lag = 20, type = "Ljung-Box",
         fitdf = sum(arimaorder(fit_T1)[c(1, 3)]))

# --Forecast and Accuracy--
fc_tfr <- forecast(fit_T1, h = 12)
fc_tlb <- forecast(fit_L1, h = 12)
accuracy(fc_tfr, tfr_test)
accuracy(fc_tlb, tlb_test)

```

## Report GitHub Repo

[https://github.com/XCalibr3S/TSA\\_EDA.git](https://github.com/XCalibr3S/TSA_EDA.git)